

Jiachi ZHANG

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EDUCATION

South China University of Technology 09/2023 – 06/2027

Qualification: Bachelor of Engineering

Major: Information Engineering

Average Score: 84.52/100

Core Courses: Signals & Systems, Analog Electronics, Digital Electronics, Digital Signal Processing, Principle of Communications, Microcomputer System and Interface Technology, Programming in C++

SUMMER STUDY PROGRAM

École Polytechnique de l'Université de Nantes, France 07/2025

Core Courses: Français, Théorie de l'Information

INTERNSHIP EXPERIENCE

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences 10/2025 – Present

Position: Visiting Research Intern | Advisors: Dr. Wenqi Fang & PhD candidate Zhuoyu Wu

- Carried out independent research by identifying research gaps, developing research hypotheses, designing experimental protocols, and conducting hands-on experiments and data analysis.
- Compiled academic materials by reviewing relevant literature and drafting concise and structured experimental summaries.
- Authored and revised the thesis, ensuring a clear, logically rigorous scientific narrative grounded in experimental results.

RESEARCH EXPERIENCE

Efficient Medical Image Segmentation via Recursive Parameter Reuse 09/2025 – Present

- Studied medical image segmentation efficiency under extreme parameter scales, and adopted recursive parameter reuse to preserve segmentation fidelity with minimal parameter consumption.
- Built a multi-granularity recursive lightweight model family, enabling the **1.3K-parameter** model to attain over **0.71 mean Dice** and **21,009× parameter efficiency** against U-Net (work under review at *ACM Multimedia 2026*).
- Designed a plug-and-play recursive refinement module with only **~0.00019M parameters** and zero static computation, raising the Dice score by over **1.3 percentage points** (work under review at *ICONIP 2026*).
- Boosted incremental parameter efficiency by **four orders of magnitude** over conventional segmentation refinement blocks, with negligible memory and latency overhead.

Deep Learning-based Glomerular Pathological Image Detection 04/2024 – 03/2025

- Designed color clustering and color transfer modules to mitigate staining variance and augment glomerular pathological image datasets.
- Constructed a YOLO11-based detection framework effectively and adopted Focal Loss to balance positive-and-negative samples in the glomerular detection task.
- Integrated EMA multi-scale attention to enhance feature fusion and suppress tissue noise; validated via ablation, achieving **94% mAP**, outperforming baseline models.

PROJECT EXPERIENCE

Macroeconomic Indicator Forecasting System via SAITS and Time-Series LLMs 07/2025 – 08/2025

- Adopted the SAITS algorithm for multivariate time-series imputation, reconstructing high-fidelity continuous feature spaces under limited labeled data.
- Integrated TimeLLM and ChatTS, designed cross-modal alignment modules to capture temporal evolution trends and enhance prediction accuracy.
- Constructed a modular training and evaluation system; **Top 10** in the E Fund FinTech Challenge.

FPGA-Based Shooting Confrontation Game 03/2026

- Developed a dual-mode FPGA shooting game supporting 1280×800@60fps HDMI output, infrared control, and real-time score display.
- Designed HDMI display drivers and SDRAM caching logic; implemented AABB-based collision detection for game objects.
- Used finite state machines to control seven game states; received the **Excellent Course Design Award**.

EXTRACURRICULAR EXPERIENCE

President, Art Troupe, Youth League Committee 11/2024 – 11/2025

- Participated in the production and singing of the department song; performed in the college band on various occasions.
- Managed departmental affairs; key organizer of the college graduation gala for multiple years.

Study Representative, Class Committee 09/2023 – 10/2025

- Collected and distributed daily assignments; completed the liaison work for academic affairs.
- Communicated actively between teachers and classmates; released official university notifications in a timely manner.

HONORS & AWARDS

- 3rd Prize Scholarship, South China University of Technology 12/2025
- Merit Student, South China University of Technology 12/2025
- Advanced Individual in Student Work, SCUT 09/2025
- Active Participant in Student Work, SCUT 09/2025
- 10th Place, Pazhou Algorithm Competition — E Fund Cup FinTech Challenge 08/2025
- Outstanding League Cadre, SCUT 05/2025

ADDITIONAL SKILLS

Language Ability: French (basic), English (fluent, can be used as a studying language)

Programming & Development: Python, PyTorch, Keil, MATLAB, Vivado